

MOHAMMAD AHSAN

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Summary

Software engineer with 8+ years of experience specializing in full-stack development using MEAN/MERN stacks. Proven expertise in building scalable web applications with React, Redux, and Node.js, following both Agile and Waterfall methodologies. Skilled in developing MVC-based applications, designing RESTful APIs, and ensuring seamless frontend-backend integration.

Strong in Object-Oriented Programming (OOP) and Responsive Web Design (RWD) using HTML5, CSS3, and JavaScript. Experienced in resolving cross-browser compatibility issues and managing databases with SQL/PL-SQL. Proficient in cloud deployments using Docker and Kubernetes on AWS and Azure, and orchestrating applications in containerized environments.

Hands-on experience with CI/CD pipelines, automated testing, and mentoring teams. Expertise in leveraging npm, Express, and Fastify for backend development. Adept at collaborating with cross-functional teams to deliver high-traffic, enterprise-level applications while promoting best practices in security, DevOps, and maintainable code.

Skill Highlights

- **Web Technologies:** HTML5, CSS3, JavaScript, ES6, AJAX, jQuery, WordPress, Typescript, JSON, Bootstrap, Angular.js, React.js, Backbone JS, Express JS, Vue.JS, Node.js, Mongoose, Python, Next.js, Tailwind CSS
- **UI/UX Designer Tools:** Figma, Adobe XD, Wireframing, Prototyping.
- **Databases:** NoSQL, MySQL Server, Mongo DB, PostgreSQL, DynamoDB, Firebase.
- **Version Controls:** GitHub, Bit Bucket, Git,
- **Cloud Platform:** AWS S3, GCP, AWS Lambda, Azure
- **Testing Tools and Packages:** Junit3.8/4.0, NPM, Jasmine, Mocha, Cucumber, Sauce Labs.
- **Containerization:** Docker, OpenShift, Kubernetes
- **CI/CD Tools:** Jenkins, CircleCI, GitLab, OAuth
- **Other Applications and frameworks:** GraphQL, Restful API, Swagger, Terraform, RabbitMQ, KAFKA, UI/UX, Data Structure, Azure SQL, Elasticsearch, ES Version, Algorithms, Unit testing, Ajax, NetBeans, Eclipse, IntelliJ IDEA.
- **Methodologies:** Agile, UML Waterfall, Scrum.

Experience

Lead Full Stack Developer || (BD) Becton, Dickinson

06/2021 to Present

- Lead a team of developers and planned technical roadmaps, set quarterly objectives, and track progress.
- Worked closely with product managers, designers, and stakeholders to understand architectural requirements and translate them into technical solutions.
- Implemented test-driven development (TDD) practices, leading to higher code quality and fewer bugs.
- Reduced page load times by 47% through optimization techniques, enhancing user experience for thousands of BD users and Integrated **payment gateway**, facilitating over a million secure transactions annually.
- Implemented circuit breakers and retry mechanisms in RESTful APIs to improve fault tolerance in distributed systems.
- Conducting AngularJS testing through frameworks such as Jasmine and Karma to conduct unit tests.
- Integrated third-party APIs and services, such as payment gateways or geolocation services using Node.js.
- I designed an intuitive GraphQL schema and implemented it using Node.js
- Set up monitoring tools like Kibana to track the health and performance of Elasticsearch clusters and leverage logging for debugging and troubleshooting.
- Implemented Kanban boards using tools like Trello or Jira to manage task backlogs, track work in progress (WIP), and visualize project timelines, facilitating agile project management.
- Designed and implemented caching strategies for RESTful APIs, leveraging mechanisms like ETag headers and caching directives to enhance performance.
- Implemented A/B testing, leading to a 11% increase in user conversions for the task posting process.
- Collaborated with the mobile app team to create a seamless cross-platform experience, resulting in a 24% increase in app retention.
- Employed Express.js middleware to manage Cross-Origin Resource Sharing (CORS), ensuring smooth communication between the AngularJS frontend and Node.js backend.
- Utilized AWS services to optimize infrastructure, resulting in a 40% decrease in operational costs.

- Optimizing data retrieval and storage patterns aligning with Lambda execution.
- Implementing Node.js function as a service (FaaS) on serverless platforms like AWS Lambda.
- Utilized PostgreSQL's extensibility to develop custom functions and extensions, addressing unique business requirements.
- Integrating Vue.js components with Web Components for seamless interoperability and ensuring smooth communication between Vue.js and Web Components.
- Utilized database monitoring and profiling tools like pg_stat_statements, MySQL Performance Schema to identify performance bottlenecks and optimize database resource utilization.
- Simplifying DynamoDB data access in GraphQL APIs using AWS AppSync effectively.
- Implemented server-side data fetching using libraries like SWR or React Query to prefetch and cache data.
- Orchestration of a microservices architecture with Kubernetes, utilizing Helm charts and Kubernetes Operators to handle complex applications, resulting in enhanced scalability and fault tolerance.
- Integrate Node.js seamlessly with AWS DynamoDB to facilitate efficient NoSQL database interactions.
- Leveraged Lambda functions to handle event-driven processing and API Gateway to manage HTTP endpoints, ensuring high availability and low latency in serverless environments.
- Involved in the transition to a CI/CD workflow, reducing deployment time by 60% and increasing feature delivery frequency.
- Developed a responsive SPA using Backbone.js with TypeScript, HTML5, CSS3, and Bootstrap for seamless user interaction. Utilized Backbone's MVC architecture to build modular, reusable UI components and improve code maintainability.
- Developed a responsive eCommerce web UI using Next.js and ReactJS, optimized for mobile, tablet, and desktop devices. Delivered high-performance, pixel-perfect interfaces aligned with digital transformation goals.
- Crafted elegant user interfaces using React.js, TypeScript, and Tailwind CSS for consistent, responsive design. Implemented server-side rendering to enhance SEO and improve page load performance.
- Developed scripts for data modeling, import/export, and DML operations using JavaScript with MongoDB, ensuring efficient data handling and integrity.
- Deployed and managed MongoDB clusters with expertise in sharding, replication, indexing, and schema design to optimize performance, scalability, and availability.
- Integrated Firebase Cloud Messaging (FCM) to deliver real-time account updates via push notifications.
- Implemented Firebase Analytics, Crashlytics, and Performance Monitoring to track user behavior and app performance insights.
- Built a Terragrunt project to maintain DRY Terraform configurations across multiple modules and automated Azure VM deployments using Terraform templates.
- Created scalable infrastructure by deploying VM scale sets and automating OpenStack resource provisioning with Python and Terraform modules.
- Developed Kafka producer and consumer programs using native clients and Spring Boot for real-time data streaming.
- Used RabbitMQ as messaging middleware for reliable microservice communication, later transitioning to AWS SQS and S3 for enhanced scalability and cloud-native integration.
- Built Single Page Applications using Angular 2 with MVVM architecture and RESTful services integration.
- Designed responsive web interfaces using Bootstrap and CSS3 media queries for optimal cross-device performance.

Senior Full Stack Developer || Dataprise

07/2015 to 05/2021

- Wrote advanced SQL queries using joins, subqueries, window functions to build reports and analytics pipelines.
- Integrate APIs and microservices to compose complete end-to-end solutions.
- Developed responsive UI across all screens, from mobile views to large screens, and cross-platform web applications using React
- Executed thorough code refactoring to harmonize Angular code patterns with React's component-based architecture.
- Enhanced application security, implementing OAuth2 authentication for protecting user data.
- By using WebAuthn for the front-end and secure credential handling on the back-end, you can create a strong passwordless authentication system. This method not only enhances security but also makes the login process simpler and more user-friendly.
- Optimized Node.js application performance using clustering, caching with Redis, profiling to identify and resolving bottlenecks.
- Pair Mocha with Chai, an assertion library, for expressive and readable assertions, facilitating the creation of comprehensive test cases.
- In one of the projects, prioritized creating a user-friendly interface for passkey registration and authentication. I ensured clear instructions were provided to guide users through each step of the process. Additionally, implemented helpful feedback mechanisms to keep users informed and confident during the passkey setup and authentication flows.
- I've utilized Swagger Editor to write OpenAPI specifications, defining endpoints, parameters, request bodies, responses, and authentication schemes.
- Orchestrated API integration tests using tools like Postman to validate API functionality and ensure compliance with specifications.

- Setting up reverse proxy configurations using tools like Nginx for Node.js applications.
- Utilized Redux for state management, resulting in a 34% reduction in UI-related bugs.
- Organize and structure tests using Mocha test suites, leveraging features like hooks for setup and teardown operations.
- Expertise in crafting efficient data models for DynamoDB's NoSQL database.
- Leveraged GraphQL subscriptions over WebSockets to enable real-time data syncing.
- Architected GraphQL schemas and resolvers to streamline communication between frontend and backend, enhancing flexibility and reducing data over-fetching.
- Developed and maintained RESTful APIs, handling requests for seamless data exchange.
- Integrated MongoDB, managing a database of thousands records for robust data storage and retrieval and facilitate GitHub integration with continuous integration tools.
- Refactored existing JavaScript into TypeScript to uncover bugs through improved compile time checking.
- Expertise in deploying Node.js applications in serverless computing environments.
- Leveraged Express middleware like body-parser for parsing incoming JSON request bodies.
- Utilized Docker and Kubernetes for containerization, enabling a 60% faster deployment process.
- Switching from Angular testing frameworks to React testing libraries such as Jest and React Testing Library. Update and optimize unit tests, integration tests, and end-to-end tests for React components.
- Developed custom stored procedures and database functions in PostgreSQL using PL/pgSQL, PL/SQL.
- Developed SOAP clients and servers, implementing robust security measures to safeguard sensitive data during transmission.
- Built a responsive eCommerce web UI using Next.js and ReactJS, optimized for mobile, tablet, and desktop. Ensured pixel-perfect design implementation and efficient frontend architecture.
- Developed Java applications and Ruby/Python scripts to monitor MongoDB health and perform ad-hoc backups using mongodump/mongorestore, ensuring data reliability and system stability.
- Automated EC2 auto-scaling with load balancers using AWS Python SDK, and managed cloud instances via AWS EC2 CLI and Management Console for efficient deployment and scalability.
- Used Terraform to write Infrastructure as Code (IaC), creating scripts to provision EC2 instances, Elastic Load Balancers, and S3 buckets for scalable AWS deployments.
- Used Kafka State Store to efficiently process and manage stateful event streams.
- Developed and consumed RESTful services using Node.js with the Express framework, while handling file uploads and manipulations for dynamic content management.

Education

Bachelor of Technology, **Computer Science** from **The University of Comilla, Bangladesh in 2015**